

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281249

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Anglese; Kurt J.

DRIVE MECHANISMS FOR SURGICAL INSTRUMENTS SUCH AS FOR USE IN ROBOTIC SURGICAL SYSTEMS

Abstract

A gearbox assembly and surgical instrument including the same. The gearbox assembly includes a drive gear including a round gear and a lead screw such that a rotational input to the round gear rotates the lead screw. A first hub is threadingly engaged about the lead screw such that rotation of the lead screw translates the first hub. A second hub is spaced-apart from the first hub and engaged with a drive rod. A compression spring is disposed between the hubs. When a force acting against the drive rod is below a threshold, the rotational input translates the first hub, compression spring, second hub, and drive rod. When the force is equal to or above the threshold, the rotational input translates the first hub and compresses the compression spring against the second hub to maintain the second hub and drive rod in position.

Inventors: Anglese; Kurt J. (Lafayette, CO)

Applicant: Covidien LP (Mansfield, MA)

Family ID: 1000008619525

Appl. No.: 19/216125

Filed: May 22, 2025

Related U.S. Application Data

parent US continuation 18211888 20230620 parent-grant-document US 12310685 child US 19216125

Publication Classification

Int. Cl.: A61B34/30 (20160101); A61B17/00 (20060101); A61B17/29 (20060101); A61B34/00 (20160101)

U.S. Cl.:

CPC **A61B34/30** (20160201); **A61B17/29** (20130101); **A61B34/71** (20160201);
A61B2017/00398 (20130101); A61B2017/2926 (20130101); A61B2034/305 (20160201)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/211,888, filed on Jun. 20, 2023, which is a continuation of U.S. patent application Ser. No. 16/260,477, filed on Jan. 29, 2019, the entire contents of each of which are hereby incorporated herein by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to surgical instruments and, more specifically, to drive mechanisms for surgical instruments such as, for example, for use in robotic surgical systems.

Background of Related Art

[0003] Robotic surgical systems are increasingly utilized in various different surgical procedures. Some robotic surgical systems include a console supporting a robotic arm. One or more different surgical instruments may be configured for use with the robotic surgical system and selectively mountable to the robotic arm. The robotic arm provides one or more inputs to the mounted surgical instrument to enable operation of the mounted surgical instrument.

[0004] The number, type, and configuration of inputs provided by the robotic arm of a robotic surgical system are constraints in the design of surgical instruments configured for use with the robotic surgical system. That is, in designing a surgical instrument compatible for mounting on and use with the robotic arm of a robotic surgical system, consideration should be taken in determining how to utilize the available inputs provided by the robotic arm to achieve the desired functionality of the surgical instrument.

SUMMARY

[0005] As used herein, the term “distal” refers to the portion that is being described which is further from a surgeon, while the term “proximal” refers to the portion that is being described which is closer to a surgeon. The terms “about,” “substantially,” and the like, as utilized herein, are meant to account for manufacturing, material, environmental, use, and/or measurement tolerances and variations. Further, to the extent consistent, any of the aspects described herein may be used in conjunction with any or all of the other aspects described herein.

[0006] Provided in accordance with aspects of the present disclosure is a gearbox assembly for a surgical instrument. The gearbox assembly includes a drive gear, a first hub, a second hub, a drive rod, and a compression spring. The drive gear includes a round gear configured to receive a rotational input and a lead screw coupled to the round gear such that the rotational input to the round gear rotates the lead screw. The first hub is threadingly engaged about the lead screw such that rotation of the lead screw translates the first hub therealong. The second hub is spaced-apart from the first hub. The drive rod is engaged with the second hub. The compression spring is disposed between the first and second hubs. When a force acting against translation of the drive rod is below a threshold, the rotational input translates the first hub to, in turn, translate the compression spring to, in turn, translate the second hub and the drive rod. When the force acting against translation of the drive rod is equal to or above the threshold, the rotational input translates the first hub to, in turn, compress the compression spring against the second hub while the second hub and drive rod are maintained in position.

[0007] In an aspect of the present disclosure, an input gear disposed in meshed engagement with

the round gear of the drive gear to provide the rotational input to the round gear.

[0008] In another aspect of the present disclosure, an external input is provided and an input shaft is operably coupled between the external input and the input gear such that rotational driving of the external input provides the rotational input to the round gear.

[0009] In another aspect of the present disclosure, the first hub is a distal hub and the second hub is a proximal hub. In such aspects, the external input may be disposed proximally of the proximal hub and the input gear may be disposed distally of the distal hub.

[0010] In yet another aspect of the present disclosure, the drive rod extends coaxially through at least one of or each of: the first hub, the second hub, the compression spring, or the drive gear.

[0011] In still another aspect of the present disclosure, the first and second hubs are coupled to one another to define a maximum distance therebetween. The compression spring is partially compressed when the first and second hubs define the maximum distance therebetween.

[0012] In still yet another aspect of the present disclosure, at least one guide bar is operably coupled between the first and second hubs. The guide bar(s) defines first and second rims configured to interact with first and second shoulders of the first and second hubs, respectively, to define the maximum distance between the first and second hubs.

[0013] In another aspect of the present disclosure, the first hub is moved towards the second hub when the compression spring is compressed against the second hub to reduce a distance between the first and second hubs.

[0014] In another aspect of the present disclosure, the drive rod is engaged with the second hub via a lock plate defining a keyhole.

[0015] Also provided in accordance with aspects of the present disclosure is a surgical instrument including a housing, a shaft extending distally from the housing, an end effector assembly extending distally from the shaft and including first and second jaw members at least the first of which is movable relative to the second from a spaced-apart position to an approximated position to grasp tissue therebetween, and a gearbox assembly disposed within the housing. The gearbox assembly may be configured similarly to any of the aspects detailed hereinabove or otherwise herein such that when a force acting against movement of at least the first jaw member towards the approximated position is below a threshold, the rotational input translates the first hub to, in turn, translate the compression spring to, in turn, translate the second hub and the drive rod to move at least the first jaw member towards the approximated position. When the force acting against movement of at least the first jaw member towards the approximated position is equal to or above the threshold, the rotational input translates the first hub to, in turn, compress the compression spring against the second hub while the second hub, the drive rod, and at least the first jaw member are maintained in position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] Various aspects and features of the present disclosure are described hereinbelow with reference to the drawings wherein like numerals designate identical or corresponding elements in each of the several views.

[0017] FIG. 1 is a perspective view of a surgical instrument provided in accordance with the present disclosure configured for mounting on a robotic arm of a robotic surgical system;

[0018] FIG. 2A is a front, perspective view of a proximal portion of the surgical instrument of FIG. 1 with an outer shell removed;

[0019] FIG. 2B is a rear, perspective view of the proximal portion of the surgical instrument of FIG. 1 with the outer shell removed;

[0020] FIG. 3 is a front, perspective view of the proximal portion of the surgical instrument of FIG.

1 with the outer shell and additional internal components removed;

[0021] FIG. **4** is a schematic illustration of an exemplary robotic surgical system configured to releasably receive the surgical instrument of FIG. **1**;

[0022] FIG. **5** is a front, perspective view of a jaw drive sub-assembly of the surgical instrument of FIG. **1**;

[0023] FIG. **6** is a rear, perspective view of the jaw drive sub-assembly of the surgical instrument of FIG. **1**;

[0024] FIG. **7** is an exploded, perspective view of the jaw drive sub-assembly of the surgical instrument of FIG. **1**;

[0025] FIG. **8** is an enlarged, front perspective view of the area of detail “8” in FIG. **7**;

[0026] FIG. **9** is an enlarged, rear perspective view of the area of detail “8” in FIG. **7**;

[0027] FIG. **10** is an exploded, perspective view illustrating a drive rod and lock plate of the jaw drive sub-assembly of the surgical instrument of FIG. **1**;

[0028] FIG. **11** is an enlarged, perspective view of the area of detail “11” in FIG. **7**;

[0029] FIG. **12** is a longitudinal, cross-sectional view taken along section line “12-12” of FIG. **1**;

[0030] FIG. **13** is a longitudinal, cross-sectional view taken along section line “13-13” of FIG. **1**;

[0031] FIG. **14** is a transverse, cross-sectional view taken along section line “14-14” of FIG. **3**;

[0032] FIG. **15** is a transverse, cross-sectional view taken along section line “15-15” of FIG. **3**;

[0033] FIG. **16** is a transverse, cross-sectional view taken along section line “16-16” of FIG. **12**;

[0034] FIG. **17** is a transverse, cross-sectional view taken along section line “17-17” of FIG. **12**;

[0035] FIG. **18** is a transverse, cross-sectional view taken along section line “18-18” of FIG. **12**;

[0036] FIG. **19** is a transverse, cross-sectional view taken along section line “19-19” of FIG. **12**;

[0037] FIG. **20** is a perspective view of a distal portion of the surgical instrument of FIG. **1** with the end effector assembly disposed in an open position;

[0038] FIG. **21** is a longitudinal, cross-sectional view of another proximal portion of the surgical instrument of FIG. **1** illustrating the jaw drive sub-assembly transitioning the end effector assembly from the open position towards a closed position;

[0039] FIG. **22** is a perspective view of the distal portion of the surgical instrument of FIG. **1** with the end effector assembly disposed in the closed position; and

[0040] FIG. **23** is a longitudinal, cross-sectional view of the proximal portion of the surgical instrument of FIG. **21** illustrating the jaw drive sub-assembly with the end effector assembly in the closed position.

DETAILED DESCRIPTION

[0041] Referring to FIGS. **1-3**, a surgical instrument **10** provided in accordance with the present disclosure generally includes a housing **20**, a shaft **30** extending distally from housing **20**, and end effector assembly **40** extending distally from shaft **30**, and a gearbox assembly **100** disposed within housing **20** and operably associated with end effector assembly **40**. Instrument **10** is detailed herein as an articulating electrosurgical forceps configured for use with a robotic surgical system, e.g., robotic surgical system **1000** (FIG. **4**). However, the aspects and features of instrument **10** provided in accordance with the present disclosure, detailed below, are equally applicable for use with other suitable surgical instruments and/or in other suitable surgical systems.

[0042] With particular reference to FIG. **1**, housing **20** of instrument **10** includes first and second body portion **22a**, **22b** and a proximal face plate **24** that cooperate to enclose gearbox assembly **100** therein. Proximal face plate **24** includes apertures defined therein through which inputs **110-140** of gearbox assembly **100** extend. A pair of latch levers **26** (only one of which is illustrated in FIG. **1**) extending outwardly from opposing sides of housing **20** and enable releasable engagement of housing **20** with a robotic arm of a surgical system, e.g., robotic surgical system **1000** (FIG. **4**). An aperture **28** defined through housing **20** permits thumbwheel **440** to extend therethrough to enable manual manipulation of thumbwheel **440** from the exterior of housing **20** to, as detailed below, permit manual opening and closing of end effector assembly **40**.

[0043] Shaft **30** of instrument **10** includes a distal segment **32**, a proximal segment **34**, and an articulating section **36** disposed between the distal and proximal segments **32**, **34**, respectively. Articulating section **36** includes one or more articulating components **37**, e.g., links, joints, etc. A plurality of articulation cables **38**, e.g., four (4) articulation cables, or other suitable actuators, extend through articulating section **36**. More specifically, articulation cables **38** are operably coupled to distal segment **32** of shaft **30** at the distal ends thereof and extend proximally from distal segment **32** of shaft **30**, through articulating section **36** of shaft **30** and proximal segment **34** of shaft **30**, and into housing **20**, wherein articulation cables **38** operably couple with an articulation sub-assembly **200** of gearbox assembly **100** to enable selective articulation of distal segment **32** (and, thus end effector assembly **40**) relative to proximal segment **34** and housing **20**, e.g., about at least two axes of articulation (yaw and pitch articulation, for example). Articulation cables **38** are arranged in a generally rectangular configuration, although other suitable configurations are also contemplated.

[0044] With respect to articulation of end effector assembly **40** relative to proximal segment **34** of shaft **30**, actuation of articulation cables **38** is effected in pairs. More specifically, in order to pitch end effector assembly **40**, the upper pair of cables **38** are actuated in a similar manner while the lower pair of cables **38** are actuated in a similar manner relative to one another but an opposite manner relative to the upper pair of cables **38**. With respect to yaw articulation, the right pair of cables **38** are actuated in a similar manner while the left pair of cables **38** are actuated in a similar manner relative to one another but an opposite manner relative to the right pair of cables **38**.

[0045] Continuing with reference to FIG. **1**, end effector assembly **40** includes first and second jaw members **42**, **44**, respectively. Each jaw member **42**, **44** includes a proximal flange portion **43a**, **45a** and a distal body portion **43b**, **45b**, respectively. Distal body portions **43b**, **45b** define opposed tissue-contacting surfaces **46**, **48**, respectively. Proximal flange portions **43a**, **45a** are pivotably coupled to one another about a pivot **50** and are operably coupled to one another via a cam-slot assembly **52** including a cam pin slidably received within cam slots defined within the proximal flange portion **43a**, **45a** of at least one of the jaw members **42**, **44**, respectively, to enable pivoting of jaw member **42** relative to jaw member **44** and distal segment **32** of shaft **30** between a spaced-apart position (e.g., an open position of end effector assembly **40**) and an approximated position (e.g. a closed position of end effector assembly **40**) for grasping tissue “T” (FIGS. **20** and **22**) between tissue-contacting surfaces **46**, **48**. As an alternative to this unilateral configuration, a bilateral configuration may be provided whereby both jaw members **42**, **44** are pivotable relative to one another and distal segment **32** of shaft **30**.

[0046] In embodiments, longitudinally-extending knife channels **49** (only knife channel **49** of jaw member **44** is illustrated; the knife channel of jaw member **42** is similarly configured) are defined through tissue-contacting surfaces **46**, **48**, respectively, of jaw members **42**, **44**. In such embodiments, a knife assembly including a knife tube **62** (FIGS. **12-15**) extending from housing **20** through shaft **30** to end effector assembly **40** and a knife blade (not shown) disposed within end effector assembly **40** between jaw members **42**, **44** is provided to enable cutting of tissue “T” (FIGS. **20** and **22**) grasped between tissue-contacting surfaces **46**, **48** of jaw members **42**, **44**, respectively. Knife tube **62** (FIGS. **12-15**) is operably coupled to a knife drive sub-assembly **300** of gearbox assembly **100** (FIGS. **2A-2B**) at a proximal end thereof to enable selective actuation thereof to, in turn, reciprocate the knife blade (not shown) between jaw members **42**, **44** to cut tissue “T” (FIGS. **20** and **22**) grasped between tissue-contacting surfaces **46**, **48**.

[0047] Referring still to FIG. **1**, a drive rod **484** is operably coupled to cam-slot assembly **52** of end effector assembly **40**, e.g., engaged with the cam pin thereof, such that longitudinal actuation of drive rod **484** pivots jaw member **42** relative to jaw member **44** between the spaced-apart and approximated positions. More specifically, urging drive rod **484** proximally pivots jaw member **42** relative to jaw member **44** towards the approximated position while urging drive rod **484** distally pivots jaw member **42** relative to jaw member **44** towards the spaced-apart position. However,

other suitable mechanisms and/or configurations for pivoting jaw member **42** relative to jaw member **44** between the spaced-apart and approximated positions in response to selective actuation of drive rod **484** are also contemplated. Drive rod **484** extends proximally from end effector assembly **40** through shaft **30** and into housing **20** wherein drive rod **484** is operably coupled with a jaw drive sub-assembly **400** of gearbox assembly **100** (FIGS. 2A-2B) to enable selective actuation of end effector assembly **40** to grasp tissue “T” (FIGS. **20** and **22**) therebetween and apply a closure force within an appropriate jaw closure force range, as detailed below.

[0048] Tissue-contacting surfaces **46**, **48** of jaw members **42**, **44**, respectively, are at least partially formed from an electrically conductive material and are energizable to different potentials to enable the conduction of electrical energy through tissue “T” (FIGS. **20** and **22**) grasped therebetween, although tissue-contacting surfaces **46**, **48** may alternatively be configured to supply any suitable energy, e.g., thermal, microwave, light, ultrasonic, ultrasound, etc., through tissue “T” (FIGS. **20** and **22**) grasped therebetween for energy-based tissue treatment. Instrument **10** defines a conductive pathway (not shown) through housing **20** and shaft **30** to end effector assembly **40** that may include lead wires, contacts, and/or electrically-conductive components to enable electrical connection of tissue-contacting surfaces **46**, **48** of jaw members **42**, **44**, respectively, to an energy source (not shown), e.g., an electrosurgical generator, for supplying energy to tissue-contacting surfaces **46**, **48** to treat, e.g., seal, tissue “T” (FIGS. **20** and **22**) grasped between tissue-contacting surfaces **46**, **48**.

[0049] With additional reference to FIGS. 2A, 2B, and 3, as noted above, gearbox assembly **100** is disposed within housing **20** and includes an articulation sub-assembly **200**, a knife drive sub-assembly **300**, and a jaw drive sub-assembly **400**. Articulation sub-assembly **200** is operably coupled between first and second inputs **110**, **120**, respectively, of gearbox assembly **100** and articulation cables **38** (FIG. 1) such that, upon receipt of appropriate inputs into first and/or second inputs **110**, **120**, articulation sub-assembly **200** manipulates cables **38** (FIG. 1) to articulate end effector assembly **40** in a desired direction, e.g., to pitch and/or yaw end effector assembly **40**.

[0050] Knife drive sub-assembly **300** is operably coupled between third input **130** of gearbox assembly **100** and knife tube **62** (FIGS. 12-15) such that, upon receipt of appropriate input into third input **130**, knife drive sub-assembly **300** manipulates knife tube **62** (FIGS. 12-15) to reciprocate the knife blade (not shown) between jaw members **42**, **44** to cut tissue “T” (FIGS. **20** and **22**) grasped between tissue-contacting surfaces **46**, **48**.

[0051] Jaw drive sub-assembly **400**, as detailed below, is operably coupled between fourth input **140** of gearbox assembly **100** and drive rod **484** such that, upon receipt of appropriate input into fourth input **140**, jaw drive sub-assembly **400** pivots jaw members **42**, **44** between the spaced-apart and approximated positions to grasp tissue “T” (FIGS. **20** and **22**) therebetween and apply a closure force within an appropriate closure force range.

[0052] Gearbox assembly **100** is configured to operably interface with a robotic surgical system **1000** (FIG. 4) when instrument **10** is mounted on robotic surgical system **1000** (FIG. 4), to enable robotic operation of gearbox assembly **100** to provide the above-detailed functionality. That is, robotic surgical system **1000** (FIG. 4) selectively provides inputs to inputs **110-140** of gearbox assembly **100** to articulate end effector assembly **40**, grasp tissue “T” (FIGS. **20** and **22**) between jaw members **42**, **44**, and/or cut tissue “T” (FIGS. **20** and **22**) grasped between jaw members **42**, **44**. However, it is also contemplated that gearbox assembly **100** be configured to interface with any other suitable surgical system, e.g., a manual surgical handle, a powered surgical handle, etc. For the purposes herein, robotic surgical system **1000** (FIG. 4) is generally described.

[0053] Turning to FIG. 4, robotic surgical system **1000** is configured for use in accordance with the present disclosure. Aspects and features of robotic surgical system **1000** not germane to the understanding of the present disclosure are omitted to avoid obscuring the aspects and features of the present disclosure in unnecessary detail.

[0054] Robotic surgical system **1000** generally includes a plurality of robot arms **1002**, **1003**; a

control device **1004**; and an operating console **1005** coupled with control device **1004**. Operating console **1005** may include a display device **1006**, which may be set up in particular to display three-dimensional images; and manual input devices **1007**, **1008**, by means of which a person, e.g., a surgeon, may be able to telemanipulate robot arms **1002**, **1003** in a first operating mode. Robotic surgical system **1000** may be configured for use on a patient **1013** lying on a patient table **1012** to be treated in a minimally invasive manner. Robotic surgical system **1000** may further include a database **1014**, in particular coupled to control device **1004**, in which are stored, for example, pre-operative data from patient **1013** and/or anatomical atlases.

[0055] Each of the robot arms **1002**, **1003** may include a plurality of members, which are connected through joints, and mounted device which may be, for example, a surgical tool “ST.” One or more of the surgical tools “ST” may be instrument **10** (FIG. 1), thus providing such functionality on a robotic surgical system **1000**.

[0056] Robot arms **1002**, **1003** may be driven by electric drives, e.g., motors, connected to control device **1004**. Control device **1004**, e.g., a computer, may be configured to activate the motors, in particular by means of a computer program, in such a way that robot arms **1002**, **1003**, and, thus, their mounted surgical tools “ST” execute a desired movement and/or function according to a corresponding input from manual input devices **1007**, **1008**, respectively. Control device **1004** may also be configured in such a way that it regulates the movement of robot arms **1002**, **1003** and/or of the motors.

[0057] With reference to FIGS. 5-7 and 12-19, jaw drive sub-assembly **400** of gearbox assembly **100** is shown generally including an input shaft **410**, an input gear **420**, a drive gear **430**, a thumbwheel **440**, a spring force assembly **450**, and a drive rod assembly **480**.

[0058] Input shaft **410** includes a proximal end portion **412** operably coupled to fourth input **140** and a distal end portion **414** having input gear **420** engaged thereon such that rotational input provided to fourth input **140** drives rotation of input shaft **410** to, thereby, drive rotation of input gear **420**. Input gear **420** is disposed in meshed engagement with round gear **432** of drive gear **430** such that rotation of input gear **420**, e.g., in response to a rotational input provided at fourth input **140**, effects rotation of drive gear **430** in an opposite direction (see FIG. 16). Thumbwheel **440** is also disposed in meshed engagement with round gear **432** of drive gear **430** such that rotation of thumbwheel **440** effects rotation of drive gear **430** in an opposite direction, thus enabling manual driving of drive gear **430** via manipulation of thumbwheel **440** (see FIG. 16).

[0059] Also referring to FIG. 11, drive gear **430**, as noted above, includes a round gear **432**. Drive gear **430** further includes a lead screw **434** fixedly engaged, e.g., monolithically formed, with round gear **432** such that rotation of round gear **432** effects similar rotation of lead screw **434**.

[0060] With additional reference to FIGS. 8 and 9, spring force assembly **450** includes a proximal hub **452**, a distal hub **454**, a compression spring **456**, and a spring washer **458**. Spring force assembly **450** further includes a pair of guide bars **470**.

[0061] Proximal and distal hubs **452**, **454** of spring force assembly **450** may be identical

[0062] components that are oriented, positioned, and/or coupled to other components differently, thus providing different functionality while reducing the number of different parts required to be manufactured. The features of proximal and distal hubs **452**, **454** are detailed below to the extent necessary to facilitate understanding of the present disclosure and, thus, although some features may be detailed with respect to only one of the proximal or distal hub **452**, **454** and the function associated therewith, similar features may be provided on the other of the proximal or distal hub **452**, **454** without the associated function. Alternatively, proximal and distal hubs **452**, **454** may be manufactured as different components.

[0063] Proximal and distal hubs **452**, **454** of spring force assembly **450** each include a body **461** defining a cavity **462** and a retainer guide **463** extending radially outwardly from opposed sides of body **461**. Each retainer guide **463** defines a trough **464** and includes a shoulder **465** extending into the respective trough **464**. Proximal and distal hubs **452**, **454** are oppositely-oriented relative to one

another such that the open ends of cavities **462** face one another and such that the shoulder **465** of each pair of retainer guides **463** of proximal and distal hubs **452**, **454** face away from one another. [0064] Proximal hub **452** further includes a transverse slot **466** defined therethrough that is configured to receive lock plate **482** of drive rod assembly **480**, as detailed below, to fix lock plate **482** and, thus, a proximal end portion of drive rod **484** relative to proximal hub **452** (see FIGS. **18** and **19**). Once engaged in this manner, drive rod **484** is locked in position coaxially disposed through proximal hub **452**, distal hub **454**, compression spring **456**, and drive gear **430**.

[0065] Distal hub **454** defines a threaded central bore **468** through body **461** thereof. Threaded central bore **468** receives lead screw **434** of drive gear **430** therethrough in threaded engagement therewith such that rotation of lead screw **434** drives translation of distal hub **454** longitudinally along lead screw **434**.

[0066] Compression spring **456** is disposed between proximal and distal hubs **452**, **454** with a proximal portion thereof disposed within cavity **461** of proximal hub **452** and a distal portion thereof disposed within cavity **461** of distal hub **462**. At least a portion of compression spring **456** is disposed about and/or configured to receive a portion of lead screw **434** of drive gear **430** therethrough. Spring washer **458** is positioned within cavity **461** of proximal hub **452** between proximal hub **452** and compression spring **456**, although other configurations are also contemplated.

[0067] Each guide bar **470** is slidably received within the troughs **464** of a corresponding pair of retainer guides **463** of proximal and distal hubs **452**, **454**. Each guide bar **470** includes a pair of spaced-apart rims **472**, **474** engaged thereon that are configured to abut shoulders **465** of the respective retainer guides **463**, thereby defining a maximum distance between proximal and distal hubs **452**, **454**. However, proximal and/or distal hubs **452**, **454** are permitted to slide along guide bars **470** towards one another, as detailed below.

[0068] Referring to FIGS. **10** and **19**, in conjunction with FIGS. **5-7**, drive rod assembly **480** includes lock plate **482** and drive rod **484**. Lock plate **482** defines a central keyhole **485** and a pair of slots **486**, e.g., arcuate slots, defined on a distal face of lock plate **482** on either side of central keyhole **485**. Lock plate **482** is configured for insertion through transverse slot **466** of proximal hub **452** and, once installed therein, portions of spring washer **458** are configured for receipt within slots **486** to secure lock plate **482** in engagement within proximal hub **452**. Spring washer **458** is maintained in position within slots **486** under the bias of compression spring **456** which, at the maximum distance between proximal and distal hubs **452**, **454** (as set by rims **472**, **474** of guide bars **470** and shoulders **465** of retainer guides **463**), is pre-compressed.

[0069] Drive rod **484**, as noted above, includes a distal end portion operably coupled to cam-slot assembly **52** of end effector assembly **40** (FIG. **1**). Drive rod **484** extends proximally through shaft **30**, housing **20**, and gearbox assembly **100** (see FIGS. **1-3**) and is engaged within lock plate **482** at a proximal end portion of drive rod **484**. More specifically, drive rod **484** defines a waist **488** towards the proximal end thereof that is configured to lock in engagement within central keyhole **485** of lock plate **482**, e.g., via longitudinal translation of drive rod **484** into central keyhole **485** until waist **488** is aligned with central keyhole **485**, followed by transverse movement of drive rod **484** relative to lock plate **482**, to thereby fix the proximal end portion of drive rod **484** relative to lock plate **482** and, thus, relative to proximal hub **452** due to the engagement of lock plate **482** within proximal hub **452**.

[0070] Turning to FIGS. **3**, **12-15**, and **17**, knife drive sub-assembly **300** includes an input shaft **310**, an input gear **320**, a central gear **330** defining external threading and internal threading, and a lead screw **340**. Input shaft **310** extends parallel and offset relative to input shaft **410** and includes a proximal end portion **312** operably coupled to third input **130** of gearbox assembly **100** (FIGS. **2A** and **2B**) and a distal end portion **314** having input gear **320** engaged thereon such that rotational input provided to third input **130** drives rotation of input shaft **310** to, thereby, drive rotation of input gear **320**. Input gear **320** is disposed in meshed engagement with the external threading of

central gear **330**. Central gear **330** is coaxial with and positioned distally of drive gear **430**.

[0071] Lead screw **340** extends through central gear **330** and is threadingly engaged with the internal threading thereof such that rotation of central gear **330**, e.g., in response to a rotational input provide to third input **130**, translates lead screw **340**. Lead screw **340** is fixedly engaged about a proximal end portion of knife tube **62** such that translation of lead screw **340** translates knife tube **62**, e.g., to thereby translate the knife blade (not shown) between jaw members **42**, **44** (FIG. **1**). Lead screw **340** and knife tube **62** are coaxially disposed about drive rod **484**.

[0072] Referring to FIGS. **5-7** and **20-23**, in use, jaw members **42**, **44** are initially disposed in the spaced-apart position (FIG. **20**) and, correspondingly, proximal and distal hubs **452**, **454** are disposed in a distal-most position such drive rod **484** is disposed in a distal-most position (see FIGS. **12-15**). Further, in this position, compression spring **456** is disposed in a least-compressed condition; although, as noted above, even in the least-compressed condition, compression spring **456** is partially compressed due to the retention of compression spring **456** in a pre-compressed configuration between proximal and distal hubs **452**, **454**.

[0073] In response to an input to close end effector assembly **40**, e.g., rotational input to fourth input **140** or a manual input to rotation wheel **440**, drive shaft **410** is rotated to thereby rotate input gear **420** which, in turn, rotates drive gear **430** such that distal hub **454** is translated proximally towards proximal hub **452** (see FIG. **21**). With reference to FIG. **21**, proximal translation of distal hub **454** urges distal hub **454** against compression spring **456**. Initially, where forces resisting approximation of jaw members **42**, **44** are below a threshold corresponding to the spring value of compression spring **456**, the closure force applied by jaw members **42**, **44** is relatively low such that the urging of distal hub **454** proximally against compression spring **456** urges compression spring **456** proximally which, in turn, urges lock plate **482** and, thus, drive rod **484** proximally to pivot jaw member **42** relative to jaw member **44** from the spaced-apart position towards the approximated position to grasp tissue “T” therebetween (FIGS. **20** and **22**).

[0074] Referring to FIGS. **5-7**, **22**, and **23**, upon further approximation of jaw members **42**, **44** to grasp tissue “T” therebetween, the forces resisting approximation of jaw members **42**, **44**, e.g., tissue “T” resisting compression, may reach the threshold and, thus the closure force applied by jaw members **42**, **44** may reach a corresponding threshold. In order to maintain the closure force applied by jaw members **42**, **44** within a closure force range such as, for example, from about 3 kg/cm.sup.2 to about 16 kg/cm.sup.2, application of further closure force by jaw members **42**, **44** is inhibited beyond this point despite further rotational input to fourth input **140**. More specifically, and with reference to FIG. **23**, once the threshold has been reached, further rotational input to fourth input **140** rotates drive shaft **410**, input gear **420**, and drive gear **430** to translate distal hub **454** further proximally into compression spring **456**. However, rather than compression spring **456** urging proximal hub **452** further proximally to continue approximation of jaw members **42**, **44** and increase the closure force applied therebetween, compression spring **456** is compressed, enabling proximal hub **452** and, thus, drive rod **484** to remain in position, thus inhibiting application of additional closure force between jaw members **42**, **44**.

[0075] Referring to FIG. **22**, with tissue “T” grasped between jaw members **42**, **44** under an appropriate closure force, energy may be supplied to jaw members **42**, **44** to treat, e.g., seal tissue “T.” Thereafter, the knife blade (not shown) may be advanced between jaw members **42**, **44** to cut the treated tissue “T.” With additional reference to FIGS. **14** and **17**, in order to advance the knife blade (not shown), a rotational input is provided to input **130** to drive rotation of input shaft **310**, input gear **320**, and central gear **330**, thereby translating lead screw **340** distally such that knife tube **62** is likewise translated distally to advance the knife blade (not shown) between jaw members **42**, **44** to cut the treated tissue “T.” Alternatively, tissue “T” may be cut without first treating the tissue “T” and/or tissue “T” may be treated without subsequent cutting.

[0076] Once tissue “T” is cut, an opposite rotation input is provided to input **130** to return the knife blade (not shown) to its initial position proximally of body portions **43b**, **45b** of jaw members **42**,

44 (see FIG. 1). Thereafter, an opposite input is provided to input **140** (or rotation wheel **440**) to return jaw members **42, 44** back towards the spaced-apart position to release the sealed and/or cut tissue.

[0077] It will be understood that various modifications may be made to the embodiments disclosed herein. Therefore, the above description should not be construed as limiting, but merely as exemplifications of various embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended thereto.

Claims

1. A gearbox assembly for a surgical instrument, comprising: a drive gear including a round gear configured to receive a rotational input and a lead screw coupled to the round gear such that the rotational input to the round gear rotates the lead screw; a first hub threadingly engaged about the lead screw such that rotation of the lead screw translates the first hub therealong; a second hub spaced-apart from the first hub; a drive rod engaged with the second hub; and a compression spring disposed between the first and second hubs, wherein, when a force acting against translation of the drive rod is below a threshold, the rotational input translates the first hub to, in turn, translate the compression spring to, in turn, translate the second hub and the drive rod, and wherein, when the force acting against translation of the drive rod is equal to or above the threshold, the rotational input translates the first hub to, in turn, compress the compression spring against the second hub while the second hub and drive rod are maintained in position.
2. The gearbox assembly according to claim 1, further comprising an input gear disposed in meshed engagement with the round gear of the drive gear to provide the rotational input to the round gear.
3. The gearbox assembly according to claim 2, further comprising an external input and an input shaft operably coupled between the external input and the input gear such that rotational driving of the external input provides the rotational input to the round gear.
4. The gearbox assembly according to claim 3, wherein the first hub is a distal hub and the second hub is a proximal hub, the external input disposed proximally of the proximal hub and the input gear disposed distally of the distal hub.
5. The gearbox assembly according to claim 1, wherein the drive rod extends coaxially through at least one of: the first hub, the second hub, the compression spring, or the drive gear.
6. The gearbox assembly according to claim 1, wherein the drive rod extends coaxially through each of: the first hub, the second hub, the compression spring, and the drive gear.
7. The gearbox assembly according to claim 1, wherein the first and second hubs are coupled to one another to define a maximum distance therebetween and wherein the compression spring is partially compressed when the first and second hubs define the maximum distance therebetween.
8. The gearbox assembly according to claim 7, further comprising at least one guide bar operably coupled between the first and second hubs, the at least one guide bar defining first and second rims configured to interact with first and second shoulders of the first and second hubs, respectively, to define the maximum distance between the first and second hubs.
9. The gearbox assembly according to claim 7, wherein the first hub is moved towards the second hub when the compression spring is compressed against the second hub to reduce a distance between the first and second hubs.
10. The gearbox assembly according to claim 1, wherein the drive rod is engaged with the second hub via a lock plate defining a keyhole.
11. A surgical instrument, comprising: a housing; a shaft extending distally from the housing; an end effector assembly extending distally from the shaft, the end effector assembly including first and second jaw members, at least the first jaw member movable relative to the second jaw member from a spaced-apart position to an approximated position to grasp tissue therebetween; and a

gearbox assembly disposed within the housing, the gearbox assembly including: a drive gear including a round gear configured to receive a rotational input and a lead screw coupled to the round gear such that the rotational input to the round gear rotates the lead screw; a first hub threadingly engaged about the lead screw such that rotation of the lead screw translates the first hub therealong; a second hub spaced-apart from the first hub; a drive rod engaged with the second hub at a proximal end portion thereof and operably coupled to the end effector assembly at a distal end portion thereof such that translation of the drive rod moves at least the first jaw member from the spaced-apart position towards the approximated position; and a compression spring disposed between the first and second hubs, wherein, when a force acting against movement of at least the first jaw member towards the approximated position is below a threshold, the rotational input translates the first hub to, in turn, translate the compression spring to, in turn, translate the second hub and the drive rod to move at least the first jaw member towards the approximated position, and wherein, when the force acting against movement of at least the first jaw member towards the approximated position is equal to or above the threshold, the rotational input translates the first hub to, in turn, compress the compression spring against the second hub while the second hub, the drive rod, and at least the first jaw member are maintained in position.

12. The surgical instrument according to claim 11, wherein the gearbox assembly further comprises an input gear disposed in meshed engagement with the round gear of the drive gear to provide the rotational input to the round gear.

13. The surgical instrument according to claim 12, wherein the gearbox assembly further comprises an external input and an input shaft operably coupled between the external input and the input gear such that rotational driving of the external input provides the rotational input to the round gear.

14. The surgical instrument according to claim 13, wherein the first hub is a distal hub and the second hub is a proximal hub, the external input disposed proximally of the proximal hub and the input gear disposed distally of the distal hub.

15. The surgical instrument according to claim 11, wherein the drive rod extends coaxially through at least one of: the first hub, the second hub, the compression spring, or the drive gear.

16. The surgical instrument according to claim 11, wherein the drive rod extends coaxially through each of: the first hub, the second hub, the compression spring, and the drive gear.

17. The surgical instrument according to claim 11, wherein the first and second hubs are coupled to one another to define a maximum distance therebetween and wherein the compression spring is partially compressed when the first and second hubs define the maximum distance therebetween.

18. The surgical instrument according to claim 17,, wherein the gearbox assembly further comprises at least one guide bar operably coupled between the first and second hubs, the at least one guide bar defining first and second rims configured to interact with first and second shoulders of the first and second hubs, respectively, to define the maximum distance between the first and second hubs.

19. The surgical instrument according to claim 17, wherein the first hub is moved towards the second hub when the compression spring is compressed against the second hub to reduce a distance between the first and second hubs.

20. The surgical instrument according to claim 11, wherein the drive rod is engaged with the second hub via a lock plate defining a keyhole.
